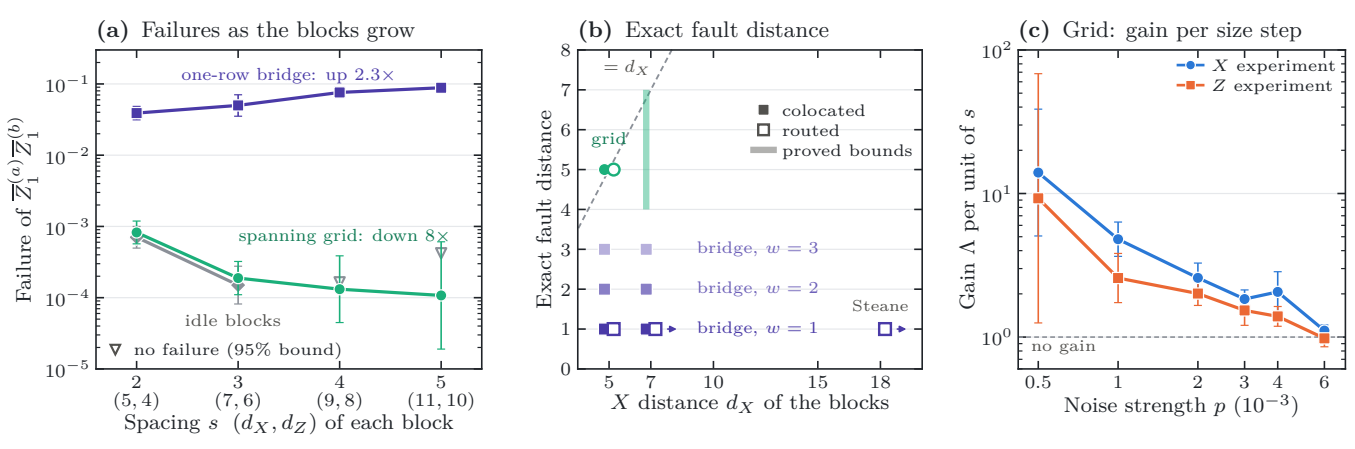

(a) Failures as the blocks grow
one-row bridge: up 2.3×
spanning grid: down 8×
idle blocks
no failure (95% bound)
Failure of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
2 (5, 4)
3 (7, 6)
4 (9, 8)
5 (11, 10)
Spacing s (d_X, d_Z) of each block
(b) Exact fault distance
$= d_X$
grid
colocated
routed
proved bounds
bridge, $w = 3$
bridge, $w = 2$
bridge, $w = 1$
Steane
Exact fault distance
0
1
2
3
4
5
6
7
8
5
7
10
15
18
X distance d_X of the blocks
(c) Grid: gain per size step
X experiment
Z experiment
Gain Λ per unit of s
10^2
10^1
10^0
no gain
0.5
1
2
3
4
6
Noise strength p (10^{-3})